

MAHARAJA P

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Pre-Final Year, B.E. Electronics & Communication Engineering | SSN College of Engineering

OBJECTIVE

Pre-final year engineering student seeking the Software Development Engineer (Intern) role at Motorq. Backend-focused builder with hands-on experience designing REST APIs, relational data models, and secure, scalable services in Java/Spring Boot — takes real ownership of problems end-to-end, cares about the quality of what ships, and uses AI coding tools to move fast.

TECHNICAL SKILLS

- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, System Design Fundamentals, SDLC
- **Languages:** Java (Core & OOP), SQL
- **Backend & APIs:** Spring Boot (Embedded Tomcat), REST API Development, JWT Authentication & Role-Based Access Control, BCrypt Encryption
- **Databases:** MySQL — Joins, ACID, Transactions, schema design
- **Tools & Deployment:** Git/GitHub, Maven, Postman, Jenkins (CI/CD), Docker (basics), AWS Cloud Practitioner (certified), Vercel, Railway, IntelliJ IDEA
- **AI-Native Workflow:** Applied GenAI — Agentic AI with Google ADK, OpenAI Whisper API integration, prompt engineering; regularly use LLMs/AI coding tools to speed up development, debugging, and prototyping

CERTIFICATIONS & EXPERIENCE

- **AWS Cloud Practitioner Essentials** — Coursera
- **Sustainability Hackathon** — Built and presented the water leak-prediction system (below); collaborated with US-based sponsors on funding and scalability feedback.
- **Zenith Hackathon** — Built and presented the Store Lens system (below) as part of a competitive team hackathon.

PROJECTS

Smart Underground Water Distribution & Leak Detection System [| GitHub](#)

Team of 2 | Backend Engineer • Java (Spring Boot), MongoDB, Python, REST APIs, IoT (ESP32)

- Built a low-cost, easy-to-operate leak-detection system for underground pipelines, aimed at villages and municipalities priced out of expensive, complex enterprise tools.
- Designed the Spring Boot backend and REST APIs to ingest live ESP32 sensor data in real time, combined with EPANET hydraulic modeling and a Python ML anomaly-detection layer to flag leaks by location.
- Applied statistical feature engineering (rolling averages, standard deviation) to cut false alarms, resolving cross-stack integration issues across Java, Python, and IoT through systematic debugging.

Store Lens — Smart Store Management System [| GitHub](#)

Team of 2 | Backend Engineer • Java, Spring Boot, MySQL, JWT, REST APIs, OpenAI Whisper API

- Built a Java/Spring Boot inventory system so any employee can instantly locate stock — solving the delays new hires faced searching for items in front of customers.
- Hand-built the backend and REST APIs end-to-end; added the OpenAI Whisper API for voice search and JWT-based role access control (AI-assisted) to block cross-shop data access.
- Automated bulk inventory updates via ACID-compliant CSV upload and added purchase-trend visibility for owners, cutting new-employee onboarding time.

LEADERSHIP & EXTRA-CURRICULAR ACTIVITIES

- **Lead Volunteer, National Service Scheme (NSS), SSN Chapter:** Led environmental cleanup drives, rural outreach, and Blood Donation/Voting Awareness camps.
- **Student Volunteer, Talent Quest for India (TQI) – Chengalpattu Cluster II:** Delivered an 8-week educational curriculum for grades 9–12, building public speaking and mentoring skills.
- **Software Coordinator, Department of ECE, SSN College of Engineering:** Mentored junior students on balancing the hardware curriculum with a software learning roadmap, and shared coding resources through peer mentorship.

ADDITIONAL INFORMATION

- **Languages:** Tamil (Native), English (Professional Proficiency)
- **Soft Skills:** Agile Teamwork, Verbal & Quantitative Reasoning, Strategic Problem-Solving, Mentoring, Working with User Stories & Estimation